

NEERAJ MAKHIJA

Senior Embedded Firmware Engineer | Cellular IoT (LTE-M / NB-IoT) · Low-Power RTOS · Secure Boot / OTA
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PROFESSIONAL SUMMARY

Senior embedded firmware engineer with 7+ years building production firmware for bare-metal and RTOS (FreeRTOS, embOS) environments on ARM Cortex-M (STM32, TI CC1314R). Owns cellular IoT firmware for a battery-powered smart gas-meter endpoint: LTE-M/NB-IoT modem subsystem architecture, AT-command and modem state-machine design, PSM duty-cycle and battery-life optimization, and secure CoAP OTA firmware download, validated with current-profiler and AT/UART traces. Strong in peripheral drivers (UART, SPI, I2C, RS232/RS485), networking/RF stacks (IPv6, DHCPv6, CoAP, Wi-SUN RF mesh), and low-level debugging (GDB, git bisect, logic analysis). Now extending into Linux device drivers, Cortex-A (AArch64), Zephyr, ESP32 BLE/Wi-Fi and RISC-V.

TECHNICAL SKILLS

Wireless & IoT	LTE-M, NB-IoT, Wi-SUN RF Mesh, BLE, Wi-Fi, Cellular AT Commands, Power Saving Mode (PSM), Battery-Life Optimization
Architectures & MCUs	ARM Cortex-M3/M4/M33, ARM Cortex-A (ARMv8-A / AArch64), RISC-V (learning), STM32, TI CC1314R, ESP32 (ESP-IDF), 8-bit MCUs
RTOS / Bare-Metal	FreeRTOS, embOS, Zephyr RTOS, NuttX, Bare-Metal
Interfaces & Drivers	UART, SPI, I2C, USB, RS232, RS485, GPIO, DMA, Interrupts, Memory-Mapped I/O
Embedded Linux	Linux Device Drivers (kernel modules, in progress), Embedded Linux (Debian/Ubuntu), POSIX Threads, Linux Shell
Networking & Protocols	TCP/IP, UDP, IPv6, DHCPv6, CoAP (RFC 7252), EAP-PSK, InterNiche
Firmware Security	Secure Boot, Secure OTA, Bootloader Development, Embedded Cryptography (AES)
Languages & Build	Embedded C, C++, Python, Bash, GCC, GNU Make, GNU ARM Toolchain, IAR
Debug, Test & Workflow	GDB, JTAG/SWD, Power Profiling (PPK2), git bisect, Logic / Protocol Analysis, Git, JIRA / TFS, Code Review

PROFESSIONAL EXPERIENCE

Senior Firmware Engineer · Landis+Gyr — Noida

May 2023 – Present

Cellular IoT (LTE-M/NB-IoT) battery endpoint for smart gas metering · Wi-SUN RF mesh for AMI smart metering

- Redesigned the cellular communication task so the modem finishes all pending work in a **single wake-up**, **cutting radio-on time per session by up to 85%** and significantly extending battery life.
- Built a **battery-life model** from real current measurements, confirming the device **comfortably exceeds its 20-year battery requirement** and letting the team see the impact of any configuration change.
- Diagnosed and fixed LTE-M/NB-IoT modem power issues using current profiling and AT command traces, including a **sleep-entry bug** that kept the modem drawing far more than its deep-sleep current.
- Implemented a **CoAP Block2 OTA firmware download client** on TI CC1314R (embOS) over UDP cellular transport with RFC 7252-compliant retransmission, Uri-Path/Uri-Query handling and flash offset management; designed fail-safe session teardown and recovery so **interrupted or aborted updates never brick the device** or resume stale sessions.
- Engineered and optimized **Wi-SUN RF mesh firmware for large-scale AMI deployments**; integrated DHCPv6, IPv6 and InterNiche TCP/IP flows (custom client-ID, solicit/advertise/request) and EAP-PSK node admission.
- Root-caused a firmware crash via **git bisect** to a GPIO configuration regression and **resolved critical field defects**, hardening production stability.

Team Lead, Software Engineer (C/C++) · Chetu Inc. — Noida

Nov 2021 – Apr 2023

- Led a **five-engineer firmware team** delivering NFC, MSR, and touchscreen POS terminal firmware; owned client-facing requirements and **shipped client projects on schedule**.
- Built **drone flight-automation** features using DJI OSDK on embedded Linux, including GPS waypoint navigation, altitude control, and multi-sensor fusion.

PROJECTS

Smart Home Voice Assistant & IoT Nodes — personal project · neerajmakhija.github.io

In progress

- Zephyr RTOS** firmware on STM32F407 Discovery driving a Nextion 4.3" touch HMI over UART; **wrote a Python serial uploader** to flash the HMI UI after the vendor SD-card update path failed.
- ESP32-WROOM-32 nodes** in C on ESP-IDF (FreeRTOS) with **BLE and Wi-Fi** for home control (relay for lighting, IR for air conditioning), linked to a PC-hosted voice pipeline; **battery-powered node** planned.

Linux Device Drivers, ARM Cortex-A & RISC-V — self-directed study

Ongoing

- Writing **Linux kernel modules and device drivers**; studying **ARMv8-A / AArch64** architecture and the **RISC-V ISA**; code and notes on the portfolio.

EARLY EXPERIENCE

R&D Embedded Engineer, Copper Connections Ltd · Delhi · Apr 2021 – Oct 2021 — PBX/telephony firmware; new call-management features; refactored legacy C across multiple device platforms, **reducing defect rates**.

Embedded Design R&D Engineer, Emtech Foundation · Delhi · Jun 2019 – Mar 2021 — End-to-end firmware for surgical equipment controllers and lab instrumentation; took a **Water Conductivity Meter from prototype to production**; firmware-level power optimization.

EDUCATION & AWARDS

B.Tech, Electronics & Communication Engineering — B.S. Anangpuria Institute of Technology & Management

2015 – 2019

Awards: Team of the Year 2024 (Landis+Gyr) · IndiaSkills 2018 — **Bronze Medal, Electronics (National)** & 1st Place (Zonal) · Faridabad Industrial Association Innovation Award